Midterm Project Proposal: SartaroshGo (Barbershop Booking App)

Course: Mobile Programming (Flutter)
Date: October 2026
Document Format: Midterm Problem & Solution Proposal (4–6 pages equivalent)
Team Name: FourD
Team Members:

1. Abdulbosit Xolikov (Team Lead & Problem/Evidence Lead)
2. Jaxongir Niyazmetov (UX/UI & Wireframes Lead)
3. Ilnur Yamaletdinov (Flutter Architecture & Navigation Lead)
4. Otabek Abdumalikov (Data Models & Technical Scope Lead)

---

1. Problem Statement & Context

1.1 The Core Problem

> **"Young working men and students in Tashkent face unpredictable 40–90 minute wait times and frequent scheduling conflicts because local neighborhood barbers rely on uncoordinated phone calls, Telegram voice messages, and informal walk-in queues."**

1.2 Target User Persona

- **Name & Age:** Jasur Karimov, 23 years old.
- **Occupation & Location:** University senior and part-time junior specialist, living in Chilanzar, Tashkent.
- **Habits & Profile:** Visits his barber every 10–14 days for a clean fade and beard trim. Uses Telegram, Payme/Click, and Yandex Go daily. Has a tight schedule balancing lectures and working hours.
- **Current Frustrations:**
    - Reaching the barber requires sending 4–5 Telegram voice notes while the barber is actively cutting another client's hair.
    - Arriving at the barbershop often reveals 3–4 walk-in patrons sitting on the waiting couch, pushing his appointment back by 45+ minutes.
    - Prices are not posted transparently; unexpected charges occur when adding beard care or styling.

- **Core Need:** An instant mobile tool to see available 30-minute time slots for his regular master, select clear services with fixed pricing in Uzbek So‘m (UZS), and reserve a confirmed slot in under 60 seconds without back-and-forth messaging.

1.3 Rubric Compliance: The 4 Tests

1. **Real:** Documented across 10 in-depth local interviews showing regular 40–90 minute delays, double-booking due to paper/napkin notes, and wasted communication overhead.
2. **Local:** Rooted in Tashkent’s neighborhood barbershop culture (*mahalla sartaroshxonalari*), heavy Telegram dependency, and local cash/card pricing habits.
3. **Narrow:** Focused solely on male grooming appointments in urban Tashkent, rather than a broad "all beauty services" platform.
4. **Solvable with a Phone:** A lightweight mobile client enables immediate schedule browsing, time-slot reservation, and booking management on-the-go.

---

2. Solution Approach

2.1 App Concept

SartaroshGo is a focused Flutter mobile app connecting urban clients with neighborhood barbers in Tashkent. It allows clients to browse nearby barbers, inspect their transparent service menus with fixed UZS prices, pick an open 30-minute time slot on an interactive calendar, and confirm appointments with instant local validation.

2.2 End-to-End User Flow (5 Steps)

1. **Browse & Select Barber:** User opens the app, views a curated list of nearby barbers (filtered by rating, district, and salon), and taps a barber card.
2. **Select Grooming Services:** User checks off desired services (e.g., *Erkaklar sochi olish - 60,000 UZS*, *Soqol tekislash - 30,000 UZS*). The sticky summary bar updates the total cost and required duration in real time.
3. **Choose Date & Time Slot:** User selects an upcoming date and taps an available green 30-minute slot (e.g., *15:30 - 16:00*). Occupied slots are clearly disabled.
4. **Enter Client Details & Confirm:** User inputs their name and phone number (`+998 XX XXX-XX-XX`) with inline validation, reviews the booking summary, and taps "Tasdiqlash" (Confirm).
5. **Booking Confirmation & Active Ticket:** A success dialog generates an active appointment card showing the countdown timer, salon address, master name, and a one-tap cancellation option.

---

3. Screen Structure & Wireframes (5 Screens)

Screen 1: Home / Barber Directory (`HomeScreen`)

- **Purpose:** Explore local barbers and view ratings, district badges, and starting prices.
- **Layout Structure:**
  - Top App Bar with search and district dropdown (`Chilanzar`, `Yunusabad`, `Mirzo Ulugbek`).
  - Horizontal category chips (`All`, `Top Rated`, `Nearest`).
  - Scrollable `ListView.builder` displaying Barber Cards.
  - Each card contains: Master photo placeholder, name, salon name, rating badge (`★ 4.9`), and "Tanlash" (Select) button.

Screen 2: Barber Profile & Service Selection (`BarberDetailScreen`)

- **Purpose:** Learn about the master's experience and select specific grooming services.
- **Layout Structure:**
  - Header banner with master details, working hours, and address.
  - Checkbox/Switch list of services (`ListView`):
    - Standard Haircut (60,000 UZS · 30 min)
    - Beard Trim & Contouring (35,000 UZS · 20 min)
    - Royal Shave & Hot Towel (45,000 UZS · 25 min)
    - Hair Wash & Styling (20,000 UZS · 15 min)
  - Sticky bottom container: Total Price counter + "Vaqtni tanlash" (Select Time) CTA button.

Screen 3: Date & Slot Picker (`BookingSlotScreen`)

- **Purpose:** Pick an available date and 30-minute time slot.
- **Layout Structure:**
  - Horizontal date strip (`DayChip` widgets: *Bugun 24 Okt*, *Ertaga 25 Okt*, etc.).
  - Morning / Afternoon / Evening section headers.
  - `GridView.builder` of time slot buttons (e.g., `10:00`, `10:30`, `11:00`, `14:00`, `17:30`).
  - Color indicators: Green (Available), Grey/Strikethrough (Booked), Blue (Selected).
  - "Davom etish" (Continue) button enabled only when a slot is active.

Screen 4: Booking Checkout & Customer Details (`CheckoutScreen`)

- **Purpose:** Collect client contact info and validate inputs before confirmation.
- **Layout Structure:**
  - `Form` with `TextFormField` for Full Name (validated: non-empty, ≥ 3 chars).
  - `TextFormField` for Phone Number with `+998` prefix formatter and regex validation.

    - Optional notes text area (e.g., "Fade 2mm on sides").
    - Price breakdown card (Subtotal + Estimated Duration).
    - Elevated action button: "Buyurtmani tasdiqlash" (Confirm Booking).

Screen 5: My Bookings & Appointment Status (`MyBookingsScreen`)

- **Purpose:** Display active and past appointments with cancellation ability.
- **Layout Structure:**
    - Segmented tabs: `Faol` (Active) and `Tarix` (Past).
    - Active booking card with high-contrast styling:
        - Date and Time prominently displayed.
        - Master name, salon address, and direct call shortcut.
        - Status chip: `Kutilmoqda` (Confirmed / Waiting).
        - Outlined button: `Bekor qilish` (Cancel appointment with confirmation dialog).

---

4. Flutter Technical Architecture (Weeks 1–7 Concepts)

4.1 Widget Hierarchy & Layout Techniques

- **Scaffold & MaterialApp:** Central structure utilizing theme data customized with clean typography and modern neutral palette.
- **Column & Row Composition:** Structured information grids for barber stats, service names, and price tags.
- **ListView.builder & GridView.builder:** Memory-efficient rendering of barber lists and dynamic time slots based on model arrays.
- **Expanded, Flexible, and Padding:** Strict constraint management preventing layout overflow (`A RenderFlex overflowed...`) across various screen densities.
- **Stack & Positioned:** Used on barber cards for floating rating badges and sticky bottom confirmation bars.

4.2 State Management Strategy

- **Approach:** Lifting State Up using Flutter's built-in `StatefulWidget` and `setState` (in line with Weeks 1–7 curriculum).
- **State Locations:**
    - **Global/Parent Level (`MyApp` or root shell):** Holds the list of active bookings (`List<BookingModel> activeBookings`).
    - **Screen Level (`BarberDetailScreen`):** Manages selected service set (`Set<String> selectedServiceIds`) to compute total price dynamically.
    - **Slot Level (`BookingSlotScreen`):** Tracks `DateTime selectedDate` and `String? selectedSlotTime`.

- **Form Level (`CheckoutScreen`):** Managed by `GlobalKey<FormState>` and `TextEditingController` instances.

4.3 Navigation & Data Passing

- **Route Structure:** Declarative Navigator routes or direct `Navigator.push` with strongly typed constructor arguments:
  - Passing `BarberModel` from `HomeScreen` to `BarberDetailScreen`.
  - Passing `BookingDraft` (Barber + Selected Services + Total) from `BarberDetailScreen` to `BookingSlotScreen`.
  - Returning confirmed `BookingModel` back via `Navigator.pop(context, newBooking)`.

4.4 Forms and Validation

- **Widgets:** `Form`, `TextFormField`, `TextEditingController`, and `InputDecoration`.
- **Validation Logic:**
  - Name validator: Checks `value.trim().length >= 3`, returns *"Ismingizni to'liq kiriting"*.
  - Phone validator: Checks Uzbekistan mobile pattern `r'^\+998[0-9]{9}$'` or 9 digits input.
  - Prevents submission and highlights fields in red if `_formKey.currentState!.validate()` fails.

---

5. Dart Data Models

```dart
// 1. Barber Master Model
class BarberModel {
  final String id;
  final String name;
  final String salonName;
  final String district;
  final double rating;
  final int reviewsCount;
  final String imageUrl;
  final List<String> availableServices;

  BarberModel({
    required this.id,
    required this.name,
    required this.salonName,
    required this.district,
    required this.rating,
    required this.reviewsCount,
    required this.imageUrl,
    required this.availableServices,
```

```
  });
}

// 2. Service Item Model
class ServiceItem {
  final String id;
  final String title;
  final int priceUzs;
  final int durationMinutes;

  ServiceItem({
    required this.id,
    required this.title,
    required this.priceUzs,
    required this.durationMinutes,
  });
}

// 3. Time Slot Model
class TimeSlot {
  final String time; // e.g. "14:30"
  final bool isAvailable;

  TimeSlot({
    required this.time,
    required this.isAvailable,
  });
}

// 4. Booking Record Model
class BookingModel {
  final String bookingId;
  final BarberModel barber;
  final List<ServiceItem> services;
  final DateTime appointmentDate;
  final String slotTime;
  final String clientName;
  final String clientPhone;
  final int totalPrice;
  final String status; // 'confirmed', 'completed', 'cancelled'

  BookingModel({
    required this.bookingId,
    required this.barber,
    required this.services,
    required this.appointmentDate,
    required this.slotTime,
    required this.clientName,
    required this.clientPhone,
    required this.totalPrice,
    this.status = 'confirmed',
```

```
  });
}
```

---

6. Out of Scope (Technical Boundary)

To remain realistic for the Weeks 1–7 Flutter syllabus, the following elements are intentionally out of scope for the midterm proposal:

1. **Remote Cloud Backend & Live WebSockets:** No Firebase/Node.js backend yet; state is simulated in-memory using Dart mock data.
2. **Online Payment Gateway Integration:** No direct Click/Payme SDK processing; payment is marked as "To'lov sartaroshxonada (naqd yoki karta)" (Pay on site).
3. **Live Geolocation & Map SDKs:** No Google Maps / Yandex MapKit widget rendering; district selection is handled via dropdowns and static tags.
4. **Push Notifications & SMS Gateways:** No automated Eskiz SMS verification codes or Firebase Cloud Messaging.

---

7. Evidence Appendix (User Interviews)

Documented from 10 one-on-one structured interviews in Tashkent (October 2026):

- **Participant 1 (Jasur M., 22, Chilanzar):** Waits 2+ hours for barber's Telegram reply; 45 min Saturday queues.
- **Participant 2 (Bekzod K., 28, Mirzo Ulugbek):** Walks into salon at 18:30, sees 4 waiting clients, loses an hour.
- **Participant 3 (Sardor A., 31, Yunusabad):** Barbers write phone bookings on paper napkins; double-booking happens frequently.
- **Participant 4 (Timur D., 20, INHA student):** Exchanging 5 Telegram audio messages just to lock down a 20-minute cut.
- **Participant 5 (Farrukh T., 26, Sergeli):** No price transparency; charged 150,000 UZS unexpectedly without prior notice.
- **Participant 6 (Shoxrux E., 24, Bank teller):** Strict 1-hour lunch break ruined because queue order was not honored.
- **Participant 7 (Otabek N., 34, Business owner):** Barber forgot phone reservation during Sunday rush.
- **Participant 8 (Doniyor R., 19, Almazar):** Campus barber packed on Friday afternoons (60–90 min wait).
- **Participant 9 (Azizbek Q., 29, Mirabad):** Salon Instagram DMs take 4–6 hours to respond.

- **Participant 10 (Kamron S., 25, Yashnabad):** When barber changes salons, clients lose track of schedule and pricing.

---

8. Team Roles & Contribution ("Who Did What")

- **Abdulbosit Xolikov (Team Lead & Problem/Evidence Lead):** Formulated problem statement, coordinated interviews, authored Section 1 & Section 7, managed document structure, and led presentation delivery.
- **Jaxongir Niyazmetov (UX/UI & Wireframes Lead):** Authored Section 2 & Section 3, designed user flow diagrams, crafted low-fidelity screen wireframes for all 5 screens, and established layout hierarchy.
- **Ilnur Yamaletdinov (Flutter Architecture Lead):** Authored Section 4 (widget hierarchy, navigation graph, state management strategy with lifting state up, and form validation).
- **Otabek Abdumalikov (Data Models & Technical Scope Lead):** Authored Section 5 (Dart classes, null safety) and Section 6 (out-of-scope boundaries and mock data architecture).

AI Tools Disclosure

- **AI Tools Used:** Claude / Gemini for structuring markdown documentation, proofreading rubric compliance, and refining Tailwind presentation templates. All technical concepts and Flutter choices were verified against course lectures.